

1. Administrative & Study Identification

Full Study Title: A Study of Tucatinib With Trastuzumab and mFOLFOX6 Versus Standard of Care Treatment in First-line HER2+ Metastatic Colorectal Cancer **Short Title/Acronym:** MOUNTAINEER-03 **Protocol Number:** SGNTUC-029 **NCT Number:** NCT05253651 **Sponsor/Company Name:** Pfizer (Seagen, a wholly owned subsidiary of Pfizer) **Trial Status:** Recruiting **Investigational Product:** Tucatinib + Trastuzumab + mFOLFOX6 **Phase of Development:** Phase III **Medical Monitor:** Pfizer Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective **Primary Objective:** To evaluate progression-free survival (PFS). **Secondary Objectives:** To evaluate overall survival (OS), confirmed objective response rate (cORR), and progression-free survival (PFS) by investigator assessment.

2.2 Background & Rationale Patients diagnosed with metastatic colorectal cancer (mCRC) have a poor prognosis with survival ranging 2-3 years. The prevalence of human epidermal growth factor receptor 2 (HER2) amplification is approximately 3-4% in mCRC and increases up to 8% in patients with KRAS/NRAS/BRAF wild-type (WT) CRC tumors. Tucatinib is a highly selective HER2-directed tyrosine kinase inhibitor that, in combination with trastuzumab, has demonstrated clinically meaningful activity in patients with chemotherapy-refractory, HER2-positive (HER2+), RAS WT mCRC in previous trials. This study is designed to investigate the efficacy and safety of first-line tucatinib in combination with trastuzumab and modified FOLFOX6 versus standard of care.

3. Study Design & Methodology

3.1 Design Framework **Study Type:** Interventional **Control Type:** Active Comparator (Standard of Care: mFOLFOX6 alone, mFOLFOX6 + bevacizumab, or mFOLFOX6 + cetuximab) **Blinding:** Open-label (Efficacy assessed by Blinded Independent Central Review [BICR]) **Randomization:** 1:1

3.2 Planned Enrollment & Duration **Target N\$:** Approximately 400 participants **Number of Sites:** Multicenter / Global (United States, Europe, Japan, etc.) **Estimated Study Start:** 2022 **Estimated Primary Completion:** 26-Jul-2029

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Histologically and/or cytologically confirmed adenocarcinoma of the colon or rectum which is locally advanced unresectable or metastatic. 2. Able to provide the most recently available formalin-fixed paraffin-embedded (FFPE) tumor tissue blocks (or freshly sectioned slides). If archival tissue is not available, a newly-obtained baseline biopsy of an accessible tumor lesion is required within 35 days prior to start of study treatment. 3. HER2+ disease as determined by a tissue based assay performed at a central laboratory. 4. Participant has rat sarcoma viral oncogene homolog wild-type (RAS WT) disease as determined by local or central testing. (Tissue sample must be analyzed within 1 year of biopsy date for central analysis). 5. Radiographically measurable disease per RECIST v1.1 with at least one site of disease that is measurable and has not been previously irradiated (or evidence of progression since previous radiation). 6. Eastern Cooperative Oncology Group (ECOG) Performance Status of 0 or 1. 7. CNS Inclusion: No evidence of brain metastases, or previously treated brain metastases which are asymptomatic based on contrast brain MRI.

4.2 Exclusion Criteria 1. Prior systemic anticancer therapy for CRC in the locally advanced unresectable or metastatic setting (with the exception of a maximum of 2 doses of mFOLFOX6). Note: May have received chemotherapy for CRC in the adjuvant setting if completed >6 months prior to enrollment. 2. Radiation therapy within 14 days prior to enrollment (or within 7 days in the setting of stereotactic radiosurgery). 3. Previous treatment with anti-HER2 therapy. 4. Ongoing Grade 3 or higher neuropathy. 5. Active or untreated gastrointestinal (GI) perforation at the time of screening.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: - Tucatinib: 300 mg twice daily - Trastuzumab: 8 mg/kg loading dose on Cycle 1 Day 1, followed by 6 mg/kg every 3 weeks - mFOLFOX6: Administered every 2 weeks. Includes oxaliplatin, leucovorin/levoleucovorin, and **fluorouracil** (ATC Code: L01BC02; Trade Names include Carac, Efudex, Adrucil, Fluoroplex, Tolak). **Route of Administration:** Oral (Tucatinib) and Intravenous (Trastuzumab, mFOLFOX6, and control biologic agents). **Duration of Treatment:** Cycles continue until disease progression, unacceptable toxicity, or patient withdrawal.

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care medications for adverse events. **Prohibited:** Prior anti-HER2 targeted therapies and concurrent investigational treatments.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures	:---
:---	:---	Screening	Day -28 to 0 Informed Consent, Medical History, Tumor tissue submission for HER2/RAS testing, Baseline Imaging
Baseline	Cycle 1 Day 1	Randomization, First Dose, Vital Signs, Cardiac Assessment	Interim Every 2 weeks Safety Labs, Efficacy Assessment/Imaging (typically Q8W), Concomitant Medications
End of Study	Follow-up	End of Treatment Visit, Survival Follow-up Tracking	

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Progression-free survival (PFS)
Measurement Method: Evaluated per RECIST v1.1 criteria by Blinded Independent Central Review (BICR) - defined as time from randomization to BICR assessment of disease progression or death from any cause.

7.2 Secondary & Exploratory Endpoints 1. Overall Survival (OS) 2. Confirmed Objective Response Rate (cORR) per RECIST v1.1 by BICR 3. Progression-free survival (PFS) per RECIST v1.1 by investigator assessment

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Surveillance and recording of adverse events, physical examination findings, vital signs, cardiac assessments, Eastern Cooperative Oncology Group (ECOG) performance status, concomitant medications, and laboratory tests. **Serious Adverse Events (SAEs):** Standard oncology protocol reporting timelines (typically 24 hours). **Data Monitoring Committee (DMC):** External Independent Data Monitoring Committee to assess safety signals.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) for primary efficacy; Safety Set for AE evaluations. **Sample Size Justification:** \$N \approx 400\$ randomized 1:1, adequately powered to detect a statistically significant improvement in PFS over standard of care. **Handling of Missing Data:** Censoring applied according to standard RECIST v1.1 guidelines for time-to-event endpoints.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: Scheduled onsite and remote clinical monitoring visits to ensure protocol adherence and data integrity. **Audit Trail:** Robust electronic case report form (eCRF) query management and data clarification workflows.

11. Study Identifiers & Governance

Primary ID: SGNTUC-029 **Posting ID:** 468008044150220766 **Registry Number:** NCT05253651 **Company/Sponsor:** Pfizer **Study Title:** MOUNTAINEER-03 **Official Regulatory Title:** An Open-label Randomized Phase 3 Study of Tucatinib in Combination With Trastuzumab and mFOLFOX6 Versus mFOLFOX6 Given With or Without Either Cetuximab or Bevacizumab as First-line Treatment for Subjects With HER2+ Metastatic Colorectal Cancer

12. Clinical Framework (Oncology Specific)

Primary Condition / Preferred UMLS Name: Metastatic Colorectal Cancer **Disease Area:** Colorectal Neoplasms **Semantic Type:** Neoplastic Process **Clinical Codes:** MeSH: D015179 | HPO: HP: 0100834 | SNOMED ID: 126837005 **Diagnosis Threshold:** HER2+ amplification and RAS Wild-Type **Medical Methodology:** Targeted Tyrosine Kinase Inhibitor (TKI) + Monoclonal Antibody + Cytotoxic Chemotherapy **Optimization Target:** First-line systemic disease control and delayed tumor progression

13. Study Procedures & Methodology

Management Pathway: First-line oncologic treatment combining HER2 blockade with standard chemotherapy backbone **Education Protocol (HCP):** Investigator meetings and site initiation visits detailing AE management (e.g., TKI-related diarrhea or hepatotoxicity) **Education Protocol (Patient):** Oral medication adherence training (tucatinib) and side effect reporting instructions **Quality Audit Mechanism:** Blinded Independent Central Review (BICR) for all radiologic imaging

14. Observation & Follow-Up Schedule

Total Duration: Continued until disease progression or end of trial (estimated completion 2029) **Onsite Visit Cadence:** Day 1 of every 14-day treatment cycle **Remote Monitoring Frequency:** Intermittent survival follow-up calls post-progression **Checklist Review Interval:** Regular tumor assessments (e.g., every 8 weeks)

15. Sign-off & Approval

Role	Name	Signature	Date		:---	:---	:---	:---
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]					
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]					Lead
Statistician	[Name]	_____	[DD-MMM-YYYY]					